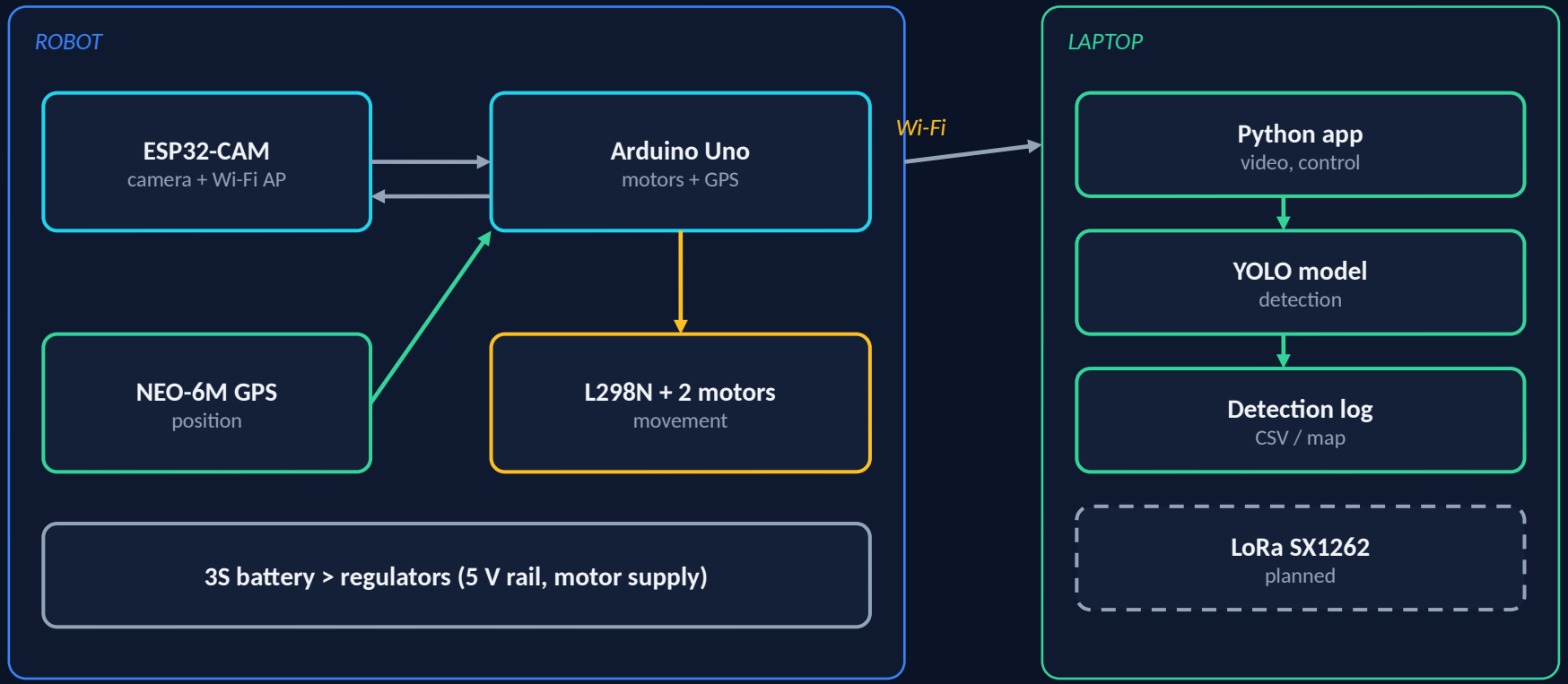
System architecture



Complete working principle

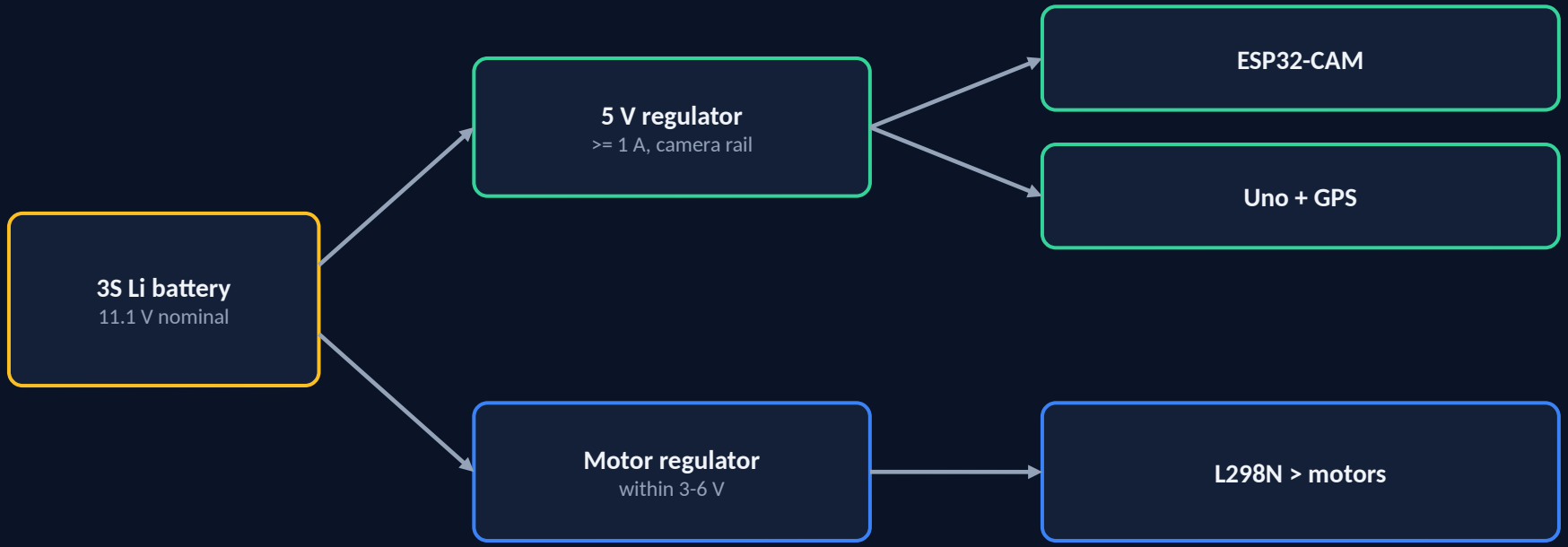


Intended workflow. Safety stop on lost video is part of the design.

Hardware and software integration

Link	Between	Carries
Wi-Fi (access point)	Laptop and ESP32-CAM	Video stream, commands
UART serial	ESP32-CAM and Uno	Drive and speed commands, GPS text
UART (software serial)	NEO-6M and Uno	NMEA position sentences
PWM + direction pins	Uno and L298N	Motor speed and direction
LoRa (planned)	Robot and base	Telemetry, detections

Battery, power distribution and chassis



Common ground across every module. Add a capacitor near the ESP32-CAM supply to absorb spikes.

Software architecture

ESP32-CAM firmware

Camera init, Wi-Fi access point, video stream.

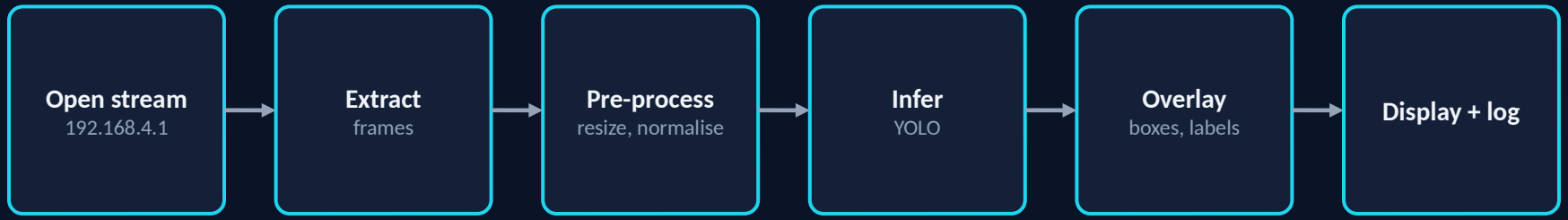
Arduino firmware

Motor direction, PWM speed, GPS parsing, serial link.

Python application

Stream receive, frames, YOLO inference, overlay, GPS association, logging.

Live video processing workflow



Frame rate and latency are to be measured; none are claimed here.

AI inference pipeline

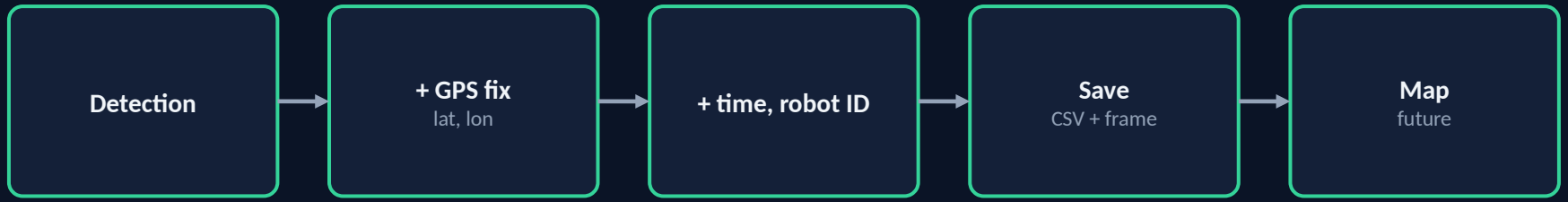


Illustrative pipeline; thresholds will be tuned during testing.

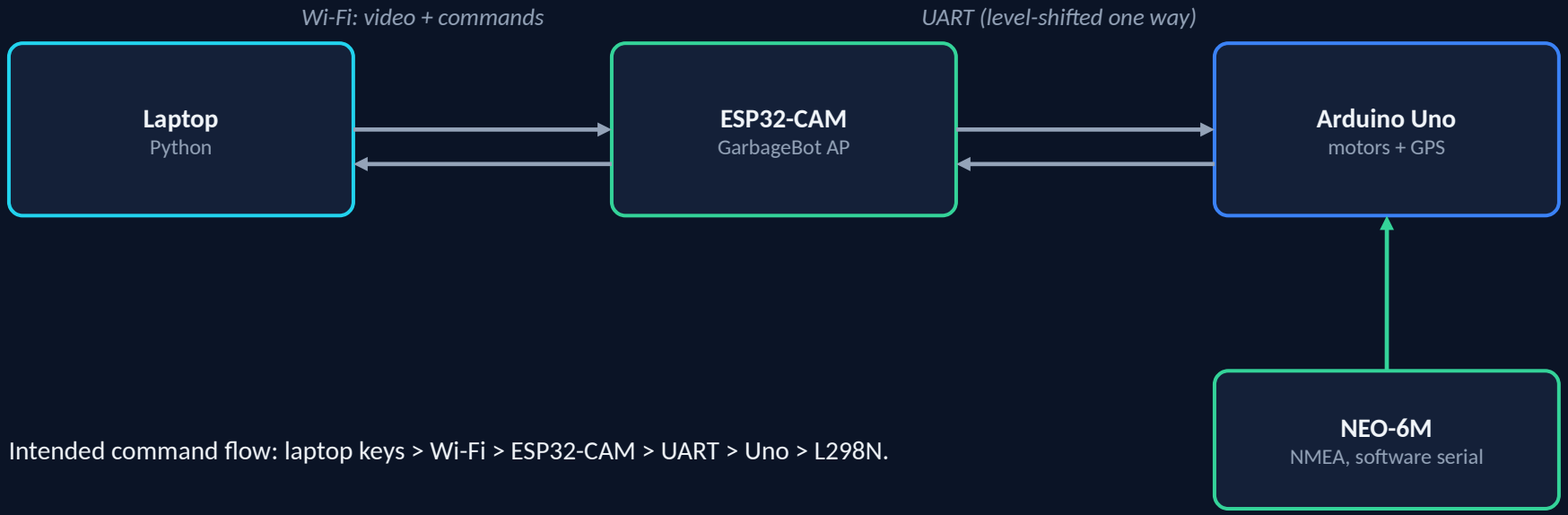
Detection records and data management

Time	Type	Conf.	Lat	Lon	Robot	Frame
10:42:07	plastic bottle	0.81	26.0000	80.0000	R1	img_001.jpg
10:42:31	paper	0.74	26.0001	80.0001	R1	img_002.jpg

ILLUSTRATIVE SAMPLE ONLY: not real measurements.



Wi-Fi and serial communication



Intended command flow: laptop keys > Wi-Fi > ESP32-CAM > UART > Uno > L298N.

End-to-end system workflow

